

# Cloud API Subsystem Design

## Table of Contents

- 1. Introduction
    - 1.1. Scope and Assumptions
  - 2. How To Read This Document
  - 3. Document Health Snapshot
  - 4. Terms and Definitions
    - 4.1. Diagram Color and Shape Legend
  - 5. Traceability Appendix
    - 5.1. Requirement Details
    - 5.2. Verification Inventory
  - 6. Reference Index
    - 6.1. Authored References
    - 6.2. Catalog References
    - 6.3. Inferred Runtime References
    - 6.4. Inferred Code References
    - 6.5. Verification References
- 

## 1. Introduction

The cloud API subsystem ingests fleet telemetry, persists metrics, and drives OTA campaigns. It is a leaf system of the connected coffee fleet, hosted on the cloud platform.

### 1.1. Scope and Assumptions

- Functional Groups and Functional Units are the stable authored design backbone.
- Runtime elements and code elements are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- Requirements and reference indexes are provided for traceability, not as primary narrative flow.

## 2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.

4. Reference Appendix: supporting reference IDs and inferred-source links.

### 3. Document Health Snapshot

| View | Authored Status | Authored Status Explanation | Units in Scope | Direct Evidence Coverage | Gap Count | Heuristic Confidence | Heuristic Basis |
|------|-----------------|-----------------------------|----------------|--------------------------|-----------|----------------------|-----------------|
|------|-----------------|-----------------------------|----------------|--------------------------|-----------|----------------------|-----------------|

### 4. Terms and Definitions

| Term                                       | Definition                                                                                                                                        |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| actor (TERM-ACTOR)                         | A person or operational role that interacts with functional units.                                                                                |
| architecture decision (TERM-DECISION)      | Authored architecture decision record with status, context, decision text, and consequences.                                                      |
| AsciiDoc diagram (TERM-ASCIIDOC-DIAGRAM)   | Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.                                 |
| AsciiDoc document (TERM-ASCIIDOC-DOCUMENT) | Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions. |
| AsciiDoc section (TERM-ASCIIDOC-SECTION)   | Structured generated document section or row model used to render authored and inferred architecture content.                                     |
| attack vector (TERM-ATTACK-VECTOR)         | A technical misuse or attack path that targets functional, referenced, or runtime elements.                                                       |
| authored mapping (TERM-AUTHORED-MAPPING)   | An explicit relationship declared in architecture inputs between authored or referenced elements.                                                 |
| catalog (TERM-CATALOG)                     | Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.                                          |
| catalog entry (TERM-CATALOG-ENTRY)         | A stable catalog term with a definition and aliases for linting, linking, and generated references.                                               |
| CLI command (TERM-CLI-COMMAND)             | Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.                      |
| cloud (FG-CLOUD)                           | Cloud ingestion and campaign control functions.                                                                                                   |
| cloud api subsystem (SYS-CLOUD-API)        | Cloud telemetry ingestion and OTA campaign control subsystem.                                                                                     |
| cloud campaign (FU-CLOUD-CAMPAIGN)         | Plans and dispatches OTA campaigns.                                                                                                               |

| Term                                             | Definition                                                                                                                                        |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| cloud ingest (FU-CLOUD-INGEST)                   | Ingests and persists fleet telemetry.                                                                                                             |
| cloud ingestion is enabled (FEAT-CLOUD-INGEST)   | Feature enabling cloud telemetry ingestion.                                                                                                       |
| code element (TERM-CODE-ELEMENT)                 | An inferred code structure or ownership element discovered from source trees and build metadata.                                                  |
| compliance mapping (TERM-COMPLIANCE-MAPPING)     | Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles. |
| control (TERM-CONTROL)                           | An authored security, policy, or operational control used to constrain behavior and reduce risk.                                                  |
| control verification (TERM-CONTROL-VERIFICATION) | Authored control verification record with method, status, threat/risk scope, findings, and evidence.                                              |
| data object (TERM-DATA-OBJECT)                   | An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.                                         |
| deployment target (TERM-DEPLOYMENT-TARGET)       | An authored deployment destination such as environment, cluster, namespace, or registry scope.                                                    |
| design document (TERM-DESIGN)                    | Authored design narrative organized by model entities and architecture views.                                                                     |
| design view (TERM-DESIGN-VIEW)                   | View-scoped design narrative attached to an authored functional group or functional unit.                                                         |
| event (TERM-EVENT)                               | An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.                                                      |
| evidence (TERM-EVIDENCE)                         | A file, artifact, or observation used to support control, risk, threat, or verification claims.                                                   |
| fleet metric (DATA-FLEET-METRIC)                 | A persisted fleet telemetry metric.                                                                                                               |
| flow (TERM-FLOW)                                 | An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.                                         |
| flow step (TERM-FLOW-STEP)                       | A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.                               |
| functional group (TERM-FUNCTIONAL-GROUP)         | A major authored capability area that groups related functional units.                                                                            |
| functional unit (TERM-FUNCTIONAL-UNIT)           | An authored working unit inside a functional group that owns specific behavior.                                                                   |

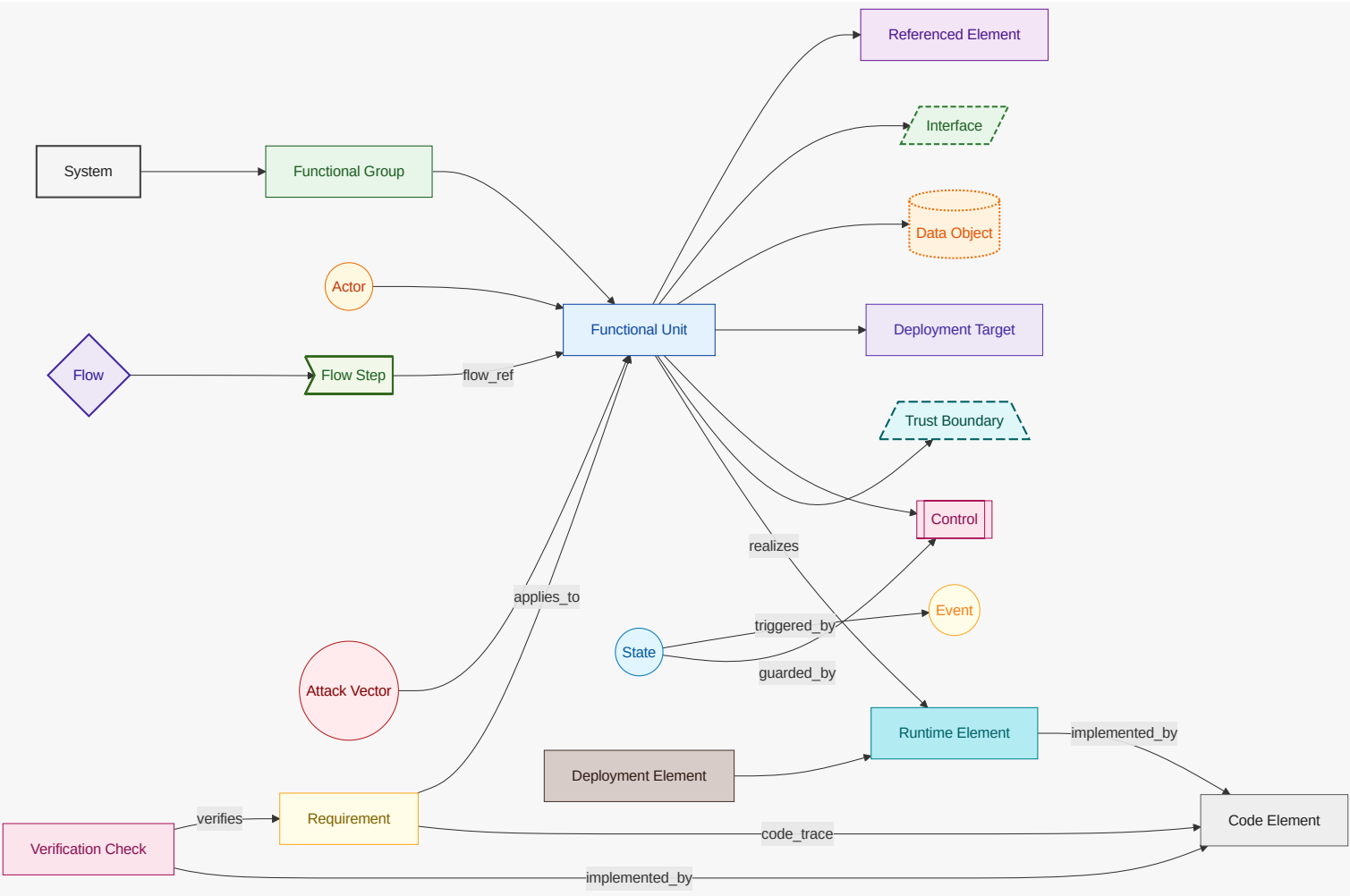
| Term                                                     | Definition                                                                                                                              |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| generation workflow (TERM-GENERATION-WORKFLOW)           | Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.                                 |
| inference hint (TERM-INFERENCE-HINT)                     | Authored source and ownership configuration used to discover runtime, code, and verification evidence.                                  |
| inferred mapping (TERM-INFERRED-MAPPING)                 | A discovered relationship that links inferred runtime/code elements upward to authored design.                                          |
| interface (TERM-INTERFACE)                               | An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.                            |
| lint run (TERM-LINT-RUN)                                 | Requirement lint configuration that controls parsing and quality checks for requirement text.                                           |
| LOBSTER activity trace (TERM-LOBSTER-ACTIVITY-TRACE)     | Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.                                |
| MCP server (TERM-MCP-SERVER)                             | Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.                            |
| MCP stdio transport (TERM-MCP-STDIO-TRANSPORT)           | Content-Length framed standard input/output transport used by the MCP command-line server.                                              |
| MCP tool (TERM-MCP-TOOL)                                 | Named MCP operation with a constrained input schema and structured response payload.                                                    |
| MCP tool response (TERM-MCP-TOOL-RESPONSE)               | Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.      |
| model (TERM-MODEL)                                       | The authored architecture model root that composes functional structure, relationships, inference hints, and views.                     |
| model bundle (TERM-BUNDLE)                               | Loaded architecture, catalog, and companion documents resolved as one working model context.                                            |
| Open OTM document (TERM-OPEN-OTM-DOCUMENT)               | Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.                     |
| OSCAL assessment results (TERM-OSCAL-ASSESSMENT-RESULTS) | Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.                          |
| OSCAL POA&M (TERM-OSCAL-POAM)                            | Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.                     |
| OSCAL SSP (TERM-OSCAL-SSP)                               | Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON. |

| Term                                                          | Definition                                                                                                                                                 |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POA&M item (TERM-POAM-ITEM)                                   | Plan of action and milestones item linked to a modeled risk and supporting artifacts.                                                                      |
| reference index (TERM-REFERENCE-INDEX)                        | Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.                                        |
| referenced element (TERM-REFERENCED-ELEMENT)                  | An architecture-relevant external, platform, or third-party dependency represented by role.                                                                |
| repository index (TERM-REPO-INDEX)                            | Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.                                         |
| requirement (TERM-REQUIREMENT)                                | A structured requirement statement that defines expected system behavior and maps to functional ownership.                                                 |
| risk (TERM-RISK)                                              | Authored risk record with likelihood, impact, response, scope, controls, and evidence.                                                                     |
| runtime element (TERM-RUNTIME-ELEMENT)                        | An inferred runtime realization element discovered from infrastructure and deployment sources.                                                             |
| security context (TERM-SECURITY-CONTEXT)                      | Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.   |
| source block (TERM-SOURCE-BLOCK)                              | Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.                                            |
| state (TERM-STATE)                                            | An authored lifecycle state used to model transition behavior, guards, and event-driven progression.                                                       |
| Structurizr DSL (TERM-STRUCTURIZR-DSL)                        | Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.                            |
| telemetry ingested event is received (EVT-TELEMETRY-INGESTED) | Fleet telemetry has been received at the cloud ingest endpoint.                                                                                            |
| threat assumption (TERM-THREAT-ASSUMPTION)                    | Authored security assumption that records scope, rationale, owner, status, and evidence.                                                                   |
| Threat Dragon document (TERM-THREAT-DRAGON-DOCUMENT)          | Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.                                                     |
| threat mitigation (TERM-THREAT-MITIGATION)                    | Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.                                           |
| threat model export (TERM-THREAT-MODEL-EXPORT)                | Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema. |

| Term                                                       | Definition                                                                                                                           |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| threat out-of-scope decision<br>(TERM-THREAT-OUT-OF-SCOPE) | Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.                        |
| threat scenario (TERM-THREAT-SCENARIO)                     | Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.                    |
| trace marker (TERM-TRACE-MARKER)                           | Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.            |
| TRLC package (TERM-TRLC-PACKAGE)                           | Generated TRLC model and requirements package used for formal requirement traceability processing.                                   |
| trust boundary (TERM-TRUST-BOUNDARY)                       | An authored trust separation zone that marks policy, identity, or security control boundaries.                                       |
| upward linking (TERM-UPWARD-LINKING)                       | Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. |
| validation (TERM-VALIDATION)                               | Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.                 |
| validation diagnostic (TERM-VALIDATION-DIAGNOSTIC)         | Structured validation finding with code, severity, message, and optional source path.                                                |
| verification check (TERM-VERIFICATION-CHECK)               | An inferred or authored verification artifact that validates requirement behavior with test evidence.                                |
| view (TERM-VIEW)                                           | Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.                             |

## 4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



# 5. Traceability Appendix

## 5.1. Requirement Details

### 5.1.1. REQ-CLOUD-001

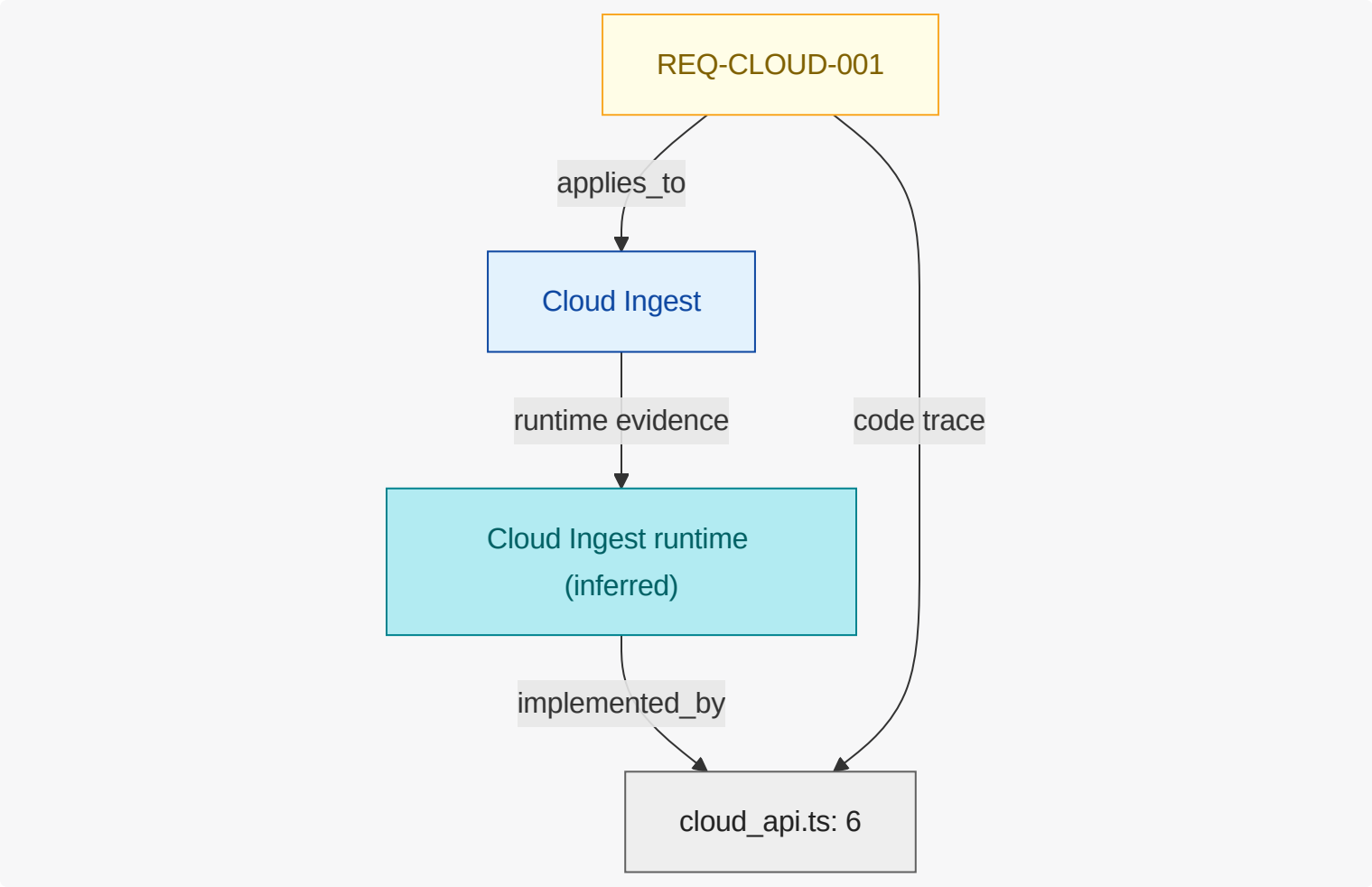
Where cloud ingestion is enabled , when telemetry ingested event is received , the cloud api subsystem shall persist a fleet metric .

NOTE

Telemetry ingestion behavior; implemented here.

### Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



### 5.1.2. REQ-CLOUD-002

Where cloud ingestion is enabled , when telemetry ingested event is received , the cloud api subsystem shall authenticate the reporting machine.

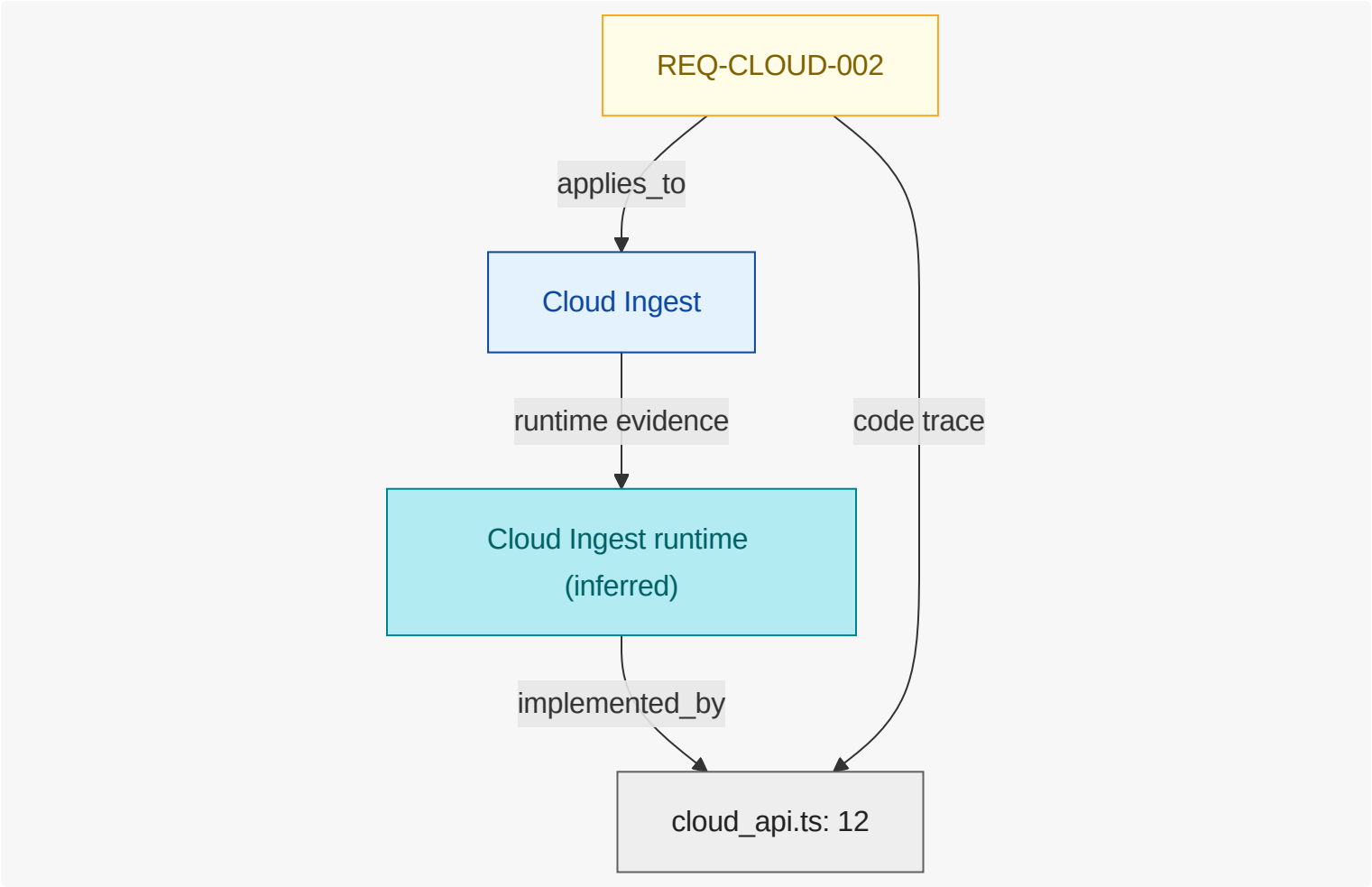
NOTE

Ingest authentication control.



Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: this requirement-to-unit mapping extended with inferred runtime/code plus verification evidence attached to each requirement and unit.



5.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

## 6. Reference Index

### 6.1. Authored References

#### CTRL-INGEST-AUTH

category=access-control; description=Authenticate machines before accepting telemetry.

| Key          | Value                 |
|--------------|-----------------------|
| Kind         | Control               |
| Name         | Ingest Authentication |
| Mentioned In | REQ-CLOUD-002         |

#### DO-FLEET-METRIC

termRef=n/a; schemaRef=n/a; sensitivity=n/a

| Key          | Value         |
|--------------|---------------|
| Kind         | Data Object   |
| Name         | Fleet Metric  |
| Mentioned In | REQ-CLOUD-001 |

#### IF-CLOUD-INGEST

protocol=https; endpoint=/ingest; schemaRef=n/a; owner=FU-CLOUD-INGEST

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Interface              |
| Name         | Cloud Ingest Interface |
| Mentioned In | n/a                    |

#### REQ-CLOUD-001

Where cloud ingestion is enabled, when telemetry ingested event is received, the cloud api subsystem shall persist a fleet metric.

| Key  | Value         |
|------|---------------|
| Kind | Requirement   |
| Name | REQ-CLOUD-001 |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

REQ-CLOUD-002

Where cloud ingestion is enabled, when telemetry ingested event is received, the cloud api subsystem shall authenticate the reporting machine.

| Key          | Value         |
|--------------|---------------|
| Kind         | Requirement   |
| Name         | REQ-CLOUD-002 |
| Mentioned In | n/a           |

6.2. Catalog References

DATA-FLEET-METRIC

A persisted fleet telemetry metric.

| Key          | Value             |
|--------------|-------------------|
| Kind         | Catalog Data Term |
| Name         | fleet metric      |
| Mentioned In | n/a               |

EVT-TELEMETRY-INGESTED

Fleet telemetry has been received at the cloud ingest endpoint.

| Key          | Value                                |
|--------------|--------------------------------------|
| Kind         | Catalog Event                        |
| Name         | telemetry ingested event is received |
| Mentioned In | REQ-CLOUD-001 , REQ-CLOUD-002        |

FEAT-CLOUD-INGEST

Feature enabling cloud telemetry ingestion.

| Key  | Value           |
|------|-----------------|
| Kind | Catalog Feature |

| Key          | Value                         |
|--------------|-------------------------------|
| Name         | cloud ingestion is enabled    |
| Mentioned In | REQ-CLOUD-001 , REQ-CLOUD-002 |

FG-CLOUD

Cloud ingestion and campaign control functions.

| Key          | Value                    |
|--------------|--------------------------|
| Kind         | Catalog Functional Group |
| Name         | cloud                    |
| Mentioned In | n/a                      |

FU-CLOUD-CAMPAIGN

Plans and dispatches OTA campaigns.

| Key          | Value                   |
|--------------|-------------------------|
| Kind         | Catalog Functional Unit |
| Name         | cloud campaign          |
| Mentioned In | n/a                     |

FU-CLOUD-INGEST

Ingests and persists fleet telemetry.

| Key          | Value                   |
|--------------|-------------------------|
| Kind         | Catalog Functional Unit |
| Name         | cloud ingest            |
| Mentioned In | n/a                     |

SYS-CLOUD-API

Cloud telemetry ingestion and OTA campaign control subsystem.

| Key  | Value               |
|------|---------------------|
| Kind | Catalog System      |
| Name | cloud api subsystem |

| Key          | Value                         |
|--------------|-------------------------------|
| Mentioned In | REQ-CLOUD-001 , REQ-CLOUD-002 |

TERM-ACTOR

A person or operational role that interacts with functional units.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | actor                  |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-DIAGRAM

Generated diagram block, usually Mermaid, that visualizes architecture relationships in the AsciiDoc publication.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | AsciiDoc diagram       |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-DOCUMENT

Human-readable generated architecture publication document assembled from authored model, inferred evidence, diagrams, references, and decisions.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | AsciiDoc document      |
| Mentioned In | n/a                    |

TERM-ASCIIDOC-SECTION

Structured generated document section or row model used to render authored and inferred architecture content.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | AsciiDoc section       |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

### TERM-ATTACK-VECTOR

A technical misuse or attack path that targets functional, referenced, or runtime elements.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | attack vector          |
| Mentioned In | n/a                    |

### TERM-AUTHORED-MAPPING

An explicit relationship declared in architecture inputs between authored or referenced elements.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | authored mapping       |
| Mentioned In | n/a                    |

### TERM-BUNDLE

Loaded architecture, catalog, and companion documents resolved as one working model context.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | model bundle           |
| Mentioned In | n/a                    |

### TERM-CATALOG

Controlled vocabulary document used to normalize model, requirement, and generated-document terminology.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | catalog                |
| Mentioned In | n/a                    |

TERM-CATALOG-ENTRY

A stable catalog term with a definition and aliases for linting, linking, and generated references.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | catalog entry          |
| Mentioned In | n/a                    |

TERM-CLI-COMMAND

Command-line entrypoint that coordinates loading, validation, generation, and file output for an engineering-model workflow.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | CLI command            |
| Mentioned In | n/a                    |

TERM-CODE-ELEMENT

An inferred code structure or ownership element discovered from source trees and build metadata.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | code element           |
| Mentioned In | n/a                    |

TERM-COMPLIANCE-MAPPING

Authored mapping that links a model control to selected compliance controls, model scope, implementation status, evidence, and responsible roles.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | compliance mapping     |
| Mentioned In | n/a                    |

TERM-CONTROL

An authored security, policy, or operational control used to constrain behavior and reduce risk.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | control                |
| Mentioned In | REQ-CLOUD-002          |

TERM-CONTROL-VERIFICATION

Authored control verification record with method, status, threat/risk scope, findings, and evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | control verification   |
| Mentioned In | n/a                    |

TERM-DATA-OBJECT

An authored data artifact or contract shape traced across flow, interface, and implementation boundaries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | data object            |
| Mentioned In | n/a                    |

TERM-DECISION

Authored architecture decision record with status, context, decision text, and consequences.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | architecture decision  |
| Mentioned In | n/a                    |

TERM-DEPLOYMENT-TARGET

An authored deployment destination such as environment, cluster, namespace, or registry scope.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |



| Key          | Value             |
|--------------|-------------------|
| Name         | deployment target |
| Mentioned In | n/a               |

TERM-DESIGN

Authored design narrative organized by model entities and architecture views.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | design document        |
| Mentioned In | n/a                    |

TERM-DESIGN-VIEW

View-scoped design narrative attached to an authored functional group or functional unit.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | design view            |
| Mentioned In | n/a                    |

TERM-EVENT

An authored trigger signal that drives transitions, flow progress, or asynchronous outcomes.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | event                  |
| Mentioned In | n/a                    |

TERM-EVIDENCE

A file, artifact, or observation used to support control, risk, threat, or verification claims.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | evidence               |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

TERM-FLOW

An authored causal interaction sequence from user/system intent to outcome, represented as ordered steps.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | flow                   |
| Mentioned In | n/a                    |

TERM-FLOW-STEP

A single authored step in a flow that captures action, data in/out, references, and normal/error/async transitions.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | flow step              |
| Mentioned In | n/a                    |

TERM-FUNCTIONAL-GROUP

A major authored capability area that groups related functional units.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | functional group       |
| Mentioned In | n/a                    |

TERM-FUNCTIONAL-UNIT

An authored working unit inside a functional group that owns specific behavior.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | functional unit        |
| Mentioned In | n/a                    |

TERM-GENERATION-WORKFLOW

Orchestrated run that combines model loading, validation, projection, rendering, and artifact emission.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | generation workflow    |
| Mentioned In | n/a                    |

TERM-INFERENCE-HINT

Authored source and ownership configuration used to discover runtime, code, and verification evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | inference hint         |
| Mentioned In | n/a                    |

TERM-INFERRED-MAPPING

A discovered relationship that links inferred runtime/code elements upward to authored design.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | inferred mapping       |
| Mentioned In | n/a                    |

TERM-INTERFACE

An authored interface boundary (for example API, channel, endpoint, or contract) owned by a functional unit.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | interface              |
| Mentioned In | n/a                    |

TERM-LINT-RUN

Requirement lint configuration that controls parsing and quality checks for requirement text.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | lint run               |
| Mentioned In | n/a                    |

TERM-LOBSTER-ACTIVITY-TRACE

Generated LOBSTER activity JSON that links verification evidence back to formal requirement identifiers.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | LOBSTER activity trace |
| Mentioned In | n/a                    |

TERM-MCP-SERVER

Model Context Protocol server that exposes engineering-model operations to AI agents through JSON-RPC tools.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP server             |
| Mentioned In | n/a                    |

TERM-MCP-STDIO-TRANSPORT

Content-Length framed standard input/output transport used by the MCP command-line server.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP stdio transport    |
| Mentioned In | n/a                    |

TERM-MCP-TOOL

Named MCP operation with a constrained input schema and structured response payload.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value    |
|--------------|----------|
| Name         | MCP tool |
| Mentioned In | n/a      |

TERM-MCP-TOOL-RESPONSE

Structured MCP tool payload that includes success state, schema version, generated timestamp, and result data or validation error.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | MCP tool response      |
| Mentioned In | n/a                    |

TERM-MODEL

The authored architecture model root that composes functional structure, relationships, inference hints, and views.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | model                  |
| Mentioned In | n/a                    |

TERM-OPEN-OTM-DOCUMENT

Open Threat Model JSON representation generated from the engineering model for interoperable threat-model exchange.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Open OTM document      |
| Mentioned In | n/a                    |

TERM-OSCAL-ASSESSMENT-RESULTS

Generated OSCAL assessment results that summarize reviewed controls, verification findings, and modeled risks.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value                    |
|--------------|--------------------------|
| Name         | OSCAL assessment results |
| Mentioned In | n/a                      |

TERM-OSCAL-POAM

Generated OSCAL plan of action and milestones that maps modeled POA&M items and related risks into compliance JSON.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | OSCAL POA&M            |
| Mentioned In | n/a                    |

TERM-OSCAL-SSP

Generated OSCAL system security plan that maps authored system metadata, components, controls, allocations, and evidence into SSP JSON.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | OSCAL SSP              |
| Mentioned In | n/a                    |

TERM-POAM-ITEM

Plan of action and milestones item linked to a modeled risk and supporting artifacts.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | POA&M item             |
| Mentioned In | n/a                    |

TERM-REFERENCE-INDEX

Generated index of authored, catalog, runtime, code, and verification references with stable anchors and backlinks.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value           |
|--------------|-----------------|
| Name         | reference index |
| Mentioned In | n/a             |

TERM-REFERENCED-ELEMENT

An architecture-relevant external, platform, or third-party dependency represented by role.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | referenced element     |
| Mentioned In | n/a                    |

TERM-REPO-INDEX

Bounded first-party source index used to answer model-aware file, requirement, control, and threat lookup queries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | repository index       |
| Mentioned In | n/a                    |

TERM-REQUIREMENT

A structured requirement statement that defines expected system behavior and maps to functional ownership.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | requirement            |
| Mentioned In | n/a                    |

TERM-RISK

Authored risk record with likelihood, impact, response, scope, controls, and evidence.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | risk                   |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

TERM-RUNTIME-ELEMENT

An inferred runtime realization element discovered from infrastructure and deployment sources.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | runtime element        |
| Mentioned In | n/a                    |

TERM-SECURITY-CONTEXT

Security-focused generated context view that groups owned functional units and shows external actors, references, flows, controls, and trust boundaries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | security context       |
| Mentioned In | n/a                    |

TERM-SOURCE-BLOCK

Stable source reference block that records file path, optional line span, kind, summary, and linked entity IDs.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | source block           |
| Mentioned In | n/a                    |

TERM-STATE

An authored lifecycle state used to model transition behavior, guards, and event-driven progression.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | state                  |



| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

TERM-STRUCTURIZR-DSL

Generated Structurizr DSL workspace that projects authored architecture, relationships, dynamic views, and deployment metadata.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Structurizr DSL        |
| Mentioned In | n/a                    |

TERM-THREAT-ASSUMPTION

Authored security assumption that records scope, rationale, owner, status, and evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat assumption      |
| Mentioned In | n/a                    |

TERM-THREAT-DRAGON-DOCUMENT

Threat Dragon JSON representation generated from the engineering model for STRIDE threat-model review.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | Threat Dragon document |
| Mentioned In | n/a                    |

TERM-THREAT-MITIGATION

Authored mitigation record that links a threat scenario to a control, effectiveness, verification, and evidence.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |
| Name | threat mitigation      |

| Key          | Value |
|--------------|-------|
| Mentioned In | n/a   |

TERM-THREAT-MODEL-EXPORT

Generated security artifact that translates authored architecture, flows, trust boundaries, threats, and mitigations into an external threat-model schema.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat model export    |
| Mentioned In | n/a                    |

TERM-THREAT-OUT-OF-SCOPE

Authored decision that excludes a threat concern from current scope with reason, owner, expiry, and evidence.

| Key          | Value                        |
|--------------|------------------------------|
| Kind         | Engineering Model Term       |
| Name         | threat out-of-scope decision |
| Mentioned In | n/a                          |

TERM-THREAT-SCENARIO

Authored threat narrative that connects attack vectors, scope, flows, controls, risks, and verification evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | threat scenario        |
| Mentioned In | n/a                    |

TERM-TRACE-MARKER

Source-level marker such as TRLC-LINKS or ENGMODEL-LINKS used to connect declarations to requirements and model entities.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value        |
|--------------|--------------|
| Name         | trace marker |
| Mentioned In | n/a          |

#### TERM-TRLC-PACKAGE

Generated TRLC model and requirements package used for formal requirement traceability processing.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | TRLC package           |
| Mentioned In | n/a                    |

#### TERM-TRUST-BOUNDARY

An authored trust separation zone that marks policy, identity, or security control boundaries.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | trust boundary         |
| Mentioned In | n/a                    |

#### TERM-UPWARD-LINKING

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | upward linking         |
| Mentioned In | n/a                    |

#### TERM-VALIDATION

Model quality gate that checks authored documents, references, relationship semantics, and requirement lint results.

| Key  | Value                  |
|------|------------------------|
| Kind | Engineering Model Term |

| Key          | Value      |
|--------------|------------|
| Name         | validation |
| Mentioned In | n/a        |

TERM-VALIDATION-DIAGNOSTIC

Structured validation finding with code, severity, message, and optional source path.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | validation diagnostic  |
| Mentioned In | n/a                    |

TERM-VERIFICATION-CHECK

An inferred or authored verification artifact that validates requirement behavior with test evidence.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | verification check     |
| Mentioned In | n/a                    |

TERM-VIEW

Authored projection configuration that selects roots, entity kinds, mappings, audience, and abstraction.

| Key          | Value                  |
|--------------|------------------------|
| Kind         | Engineering Model Term |
| Name         | view                   |
| Mentioned In | n/a                    |

6.3. Inferred Runtime References

6.4. Inferred Code References

cloud\_api.ts

ingests and persists authenticated fleet telemetry

| Key          | Value            |
|--------------|------------------|
| Kind         | Code source_file |
| Owner        | FU-CLOUD-INGEST  |
| Source       | cloud_api.ts     |
| Mentioned In | n/a              |

cloud\_api.ts:12

Inferred code symbol owned by FU-CLOUD-INGEST from cloud\_api.ts:12.

| Key          | Value           |
|--------------|-----------------|
| Kind         | Code symbol     |
| Owner        | FU-CLOUD-INGEST |
| Source       | cloud_api.ts:12 |
| Mentioned In | n/a             |

cloud\_api.ts:6

Inferred code symbol owned by FU-CLOUD-INGEST from cloud\_api.ts:6.

| Key          | Value           |
|--------------|-----------------|
| Kind         | Code symbol     |
| Owner        | FU-CLOUD-INGEST |
| Source       | cloud_api.ts:6  |
| Mentioned In | n/a             |

## 6.5. Verification References